Mapped Tensor-Product Quadrature on the Simplex via Stick-Breaking

Sachin Natesh

1 Reference simplex and Dirichlet–Jacobi weight

Let

$$\Delta_d = \left\{ x \in \mathbb{R}^d : \; x_i \geq 0, \; \sum_{i=1}^{d} x_i \leq 1 \right\} \tag{1}$$

be the reference d-simplex. We use the Dirichlet-type Jacobi weight

$$w_\kappa(x) := C_\kappa \left(\prod_{i=1}^{d} x_i^{\kappa_i - \frac{1}{2}} \right) \left(1 - \sum_{i=1}^{d} x_i \right)^{\kappa_{d+1} - \frac{1}{2}}, \tag{2}$$

where $\kappa \in \mathbb{R}^{d+1}$ and the parameters are chosen so that the weight is integrable. The normalization constant C_κ is chosen so that

$$\int_{\Delta_d} w_\kappa(x)\, dx = 1. \tag{3}$$

2 Stick-breaking map and Jacobian

Introduce variables

$$t = (t_1, \ldots, t_d) \in [0,1]^d \tag{4}$$

and define the stick-breaking map

$$x_1 = t_1, \tag{5}$$
$$x_2 = (1 - t_1) t_2, \tag{6}$$
$$x_3 = (1 - t_1)(1 - t_2) t_3, \tag{7}$$
$$\vdots \tag{8}$$
$$x_d = \left(\prod_{j=1}^{d-1} (1 - t_j) \right) t_d. \tag{9}$$

Then $x \in \Delta_d$, and the remaining barycentric coordinate is

$$x_{d+1} := 1 - \sum_{i=1}^{d} x_i = \prod_{j=1}^{d} (1 - t_j). \tag{10}$$

The Jacobian determinant of the transformation is

$$\left| \frac{\partial x}{\partial t} \right| = \prod_{j=1}^{d} (1 - t_j)^{d-j}. \tag{11}$$

3 Factorization of the weighted measure

Substituting the stick-breaking map into the weight and multiplying by the Jacobian gives a product measure on the cube:

$$w_\kappa(x(t)) \left| \frac{\partial x}{\partial t} \right| = C_\kappa \prod_{j=1}^{d} t_j^{\kappa_j - \frac{1}{2}} (1 - t_j)^{\eta_j - \frac{1}{2}}, \tag{12}$$

where

$$\eta_j := \kappa_{j+1} + \kappa_{j+2} + \cdots + \kappa_{d+1} + \frac{d-j}{2}. \tag{13}$$

Equivalently, if the normalization is absorbed into the one-dimensional factors, the simplex integral becomes

$$\int_{\Delta_d} f(x) w_\kappa(x)\, dx = \int_{[0,1]^d} f(x(t)) \prod_{j=1}^{d} \widetilde{w}_j(t_j)\, dt, \tag{14}$$

with

$$\widetilde{w}_j(t) \propto t^{\kappa_j - \frac{1}{2}} (1 - t)^{\eta_j - \frac{1}{2}}. \tag{15}$$

4 Mapped tensor-product quadrature

For each coordinate $j \in \{1, \ldots, d\}$, let

$$\left\{ \left(t_q^{(j)}, \omega_q^{(j)} \right) \right\}_{q=1}^{Q_j} \tag{16}$$

be a Gauss–Jacobi quadrature rule on $[0, 1]$ for the weight

$$t^{\kappa_j - \frac{1}{2}} (1 - t)^{\eta_j - \frac{1}{2}}. \tag{17}$$

Taking the tensor product of these one-dimensional rules gives cube nodes

$$t^{(q)} = \left(t_{q_1}^{(1)}, \ldots, t_{q_d}^{(d)} \right) \tag{18}$$

and tensor-product weights

$$\Omega^{(q)} = \prod_{j=1}^{d} \omega_{q_j}^{(j)}. \tag{19}$$

Each cube node is mapped to the simplex by (9):

$$x^{(q)} = x\big(t^{(q)}\big). \tag{20}$$

The resulting mapped tensor-product rule is

$$\int_{\Delta_d} f(x) w_\kappa(x)\, dx \approx \sum_q \Omega^{(q)} f\big(x^{(q)}\big), \tag{21}$$

with any overall normalization handled consistently by the one-dimensional Jacobi weights.

This construction converts a weighted integral on the simplex into a tensor product of one-dimensional Jacobi-weighted integrals on $[0, 1]$, while the stick-breaking map supplies the corresponding simplex nodes.